

Assignment-3
AI 24304 Machine Learning Techniques Lab
Deadline: 21st August, 2026, 4:00 pm

Question 1:

Use the following dataset to find a simple linear regressor ($y = \beta_0 + \beta_1 x$) with respect to each feature (consider features with only continuous values) in the dataset.

<https://archive.ics.uci.edu/dataset/186/wine+quality>

The dataset has 11 features which can be used to check the quality of wine.

- a) Estimate the values of β_0 and β_1 under individual features for gradient update.
 - i. When ridge regression ($\lambda ||\beta||_2^2$) is used along with the loss term ($||\hat{y} - y||_p^p$) (where $p = 2$ when MSE is used as the loss function and $p=1$ when MAE is used as the loss function).
 - ii. When LASSO ($\lambda ||\beta||_1$) regression is used along with the loss term ($||\hat{y} - y||_p^p$) (where $p = 2$ when MSE is used as the loss function and $p = 1$ when MAE is used as the loss function).
- b) Create a plot of iterative changes in β_0 (x-axis) and β_1 (y-axis) for 100 iterations with z-axis denoting the error convergence (use both mean squared error (MSE) and mean absolute error (MAE) to record the iterative changes). Also record your iterative response in a csv file.
- c) Study the impact of the regularization hyper-parameter (λ) on the optimized solution. Find out the most optimal value of λ by performing model training under different values.
- d) Draw your observations.

Note: In-built functions for finding the gradient update for linear regressor and regularization not allowed.

Questions 2:

Repeat Question 1 under multiple linear regression ($y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 \cdots \beta_n x_n$) when all the features are used to estimate the target variable y . Individual plots are not required as all features are used at the same time. Record the gradient updates for all parameters in a csv file.

Note: In-built functions for finding the gradient update for linear regressor and regularization not allowed.

Question 3:

Use the dataset: <https://archive.ics.uci.edu/dataset/45/heart+disease>

- a) Write down a program to perform Naïve Bayesian classification
- b) Find out the confusion matrix and classification metrics (accuracy, precision, recall, specificity, true positive rate and false positive rate).
- c) Come up with the ROC/AUC curve.

Note: Direct usage of classifier function is not allowed. However, you can find your posterior distribution as a function of likelihood term and prior distribution. The features are numerical or categorical (both nominal and ordinal)

Instructions:

- a) All the programs must be submitted on google classroom **"ONLY"** within the specified deadline. Kindly stick to the deadline period as no assignment will be accepted beyond the deadline period.
- b) The file names should be same as the question number of the programs. For example, if there are five questions in "Lab Assignment x", name your files as assx-1.py, assx-2.py, assx-3.py, etc; so on and so forth; where x is the assignment number.
- c) All the above files must be wrapped into a zipped folder; with your roll number as its name in a single zipped file (.zip or .rar format). For example, if your roll number is BTECH/15004/24, name your file as "btech1500424-assx.zip" or " btech150042422-assx.rar"; where x is the assignment number.
- d) Always keep a copy of your programs stored somewhere so that it can be presented whenever asked for.
- e) Some questions may ask to note down your observations. You can prepare a PDF document for that purpose and add the same as 'obs-assx.pdf'. You can demarcate your observations for different programming questions within the PDF file.
- f) All the programs will be checked for software similarity with a tolerance limit of 30%.